

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal | Lecture 43

Histology of the Excretory Part of the Renal System

Dr. Abduraheem Farah

afarah1@sgu.edu

*Department of Anatomical Sciences
School of Medicine, St. George's University*

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Objectives

SOM.MK.1.BPM1.1.CPR.3.HCB.0542	State the functions of the proximal straight tubule.
SOM.MK.1.BPM1.1.CPR.3.HCB.0543	Describe the microscopic structure of the proximal straight tubule.
SOM.MK.1.BPM1.1.CPR.3.HCB.0544	Give the function(s) of the thin segment of descending and ascending loops of Henle.
SOM.MK.1.BPM1.1.CPR.3.HCB.0545	Describe the microscopic structure of the thin segment of descending and ascending loops of Henle.
SOM.MK.1.BPM1.1.CPR.3.HCB.0546	Give the function of the vasa rectae.
SOM.MK.1.BPM1.1.CPR.3.HCB.0547	Explain the functional significance of the vasa rectae.
SOM.MK.1.BPM1.1.CPR.3.HCB.0548	Give the function of the distal straight tubule.
SOM.MK.1.BPM1.1.CPR.3.HCB.0549	Describe the microscopic structure of the distal straight tubule.
SOM.MK.1.BPM1.1.CPR.3.HCB.0550	Give the function of the distal convoluted tubule.
SOM.MK.1.BPM1.1.CPR.3.HCB.0551	Describe the microscopic structure of the distal convoluted tubule.
SOM.MK.1.BPM1.1.CPR.3.HCB.0552	Explain the clinical significance of each component of the juxtaglomerular apparatus.
SOM.MK.1.BPM1.1.CPR.3.HCB.0553	Identify the location of the juxtaglomerular apparatus.
SOM.MK.1.BPM1.1.CPR.3.HCB.0554	Describe the site of production of renin and its role in the renin-angiotensin-aldosterone system.
SOM.MK.1.BPM1.1.CPR.3.HCB.0555	Give the function of connecting tubules.
SOM.MK.1.BPM1.1.CPR.3.HCB.0556	Describe the microscopic structure of connecting tubules.
SOM.MK.1.BPM1.1.CPR.3.HCB.0557	Distinguish between the cortical and medullary collecting ducts based on their microscopic structure, function and location.
SOM.MK.1.BPM1.1.CPR.3.HCB.0558	Describe the microscopic structure of the papillary ducts (of Bellini).
SOM.MK.1.BPM1.1.CPR.3.HCB.0559	State the functional significance of the papillary ducts

Objectives

SOM.MK.1.BPM1.1.CPR.3.HCB.0560	Describe the contents of renal interstitium.
SOM.MK.1.BPM1.1.CPR.3.HCB.0561	Describe the structure and function of minor and major calyces.
SOM.MK.1.BPM1.1.CPR.3.HCB.0562	Explain the significance of the junctional complexes of the urothelium.
SOM.MK.1.BPM1.1.CPR.3.HCB.0563	Describe the structure and function of the ureter.
SOM.MK.1.BPM1.1.CPR.3.HCB.0564	Describe the structure and function of the urinary bladder.
SOM.MK.1.BPM1.1.CPR.3.HCB.0565	Compare the male and female urethra based on length, location and epithelial lining.
SOM.MK.1.BPM1.1.CPR.3.HCB.0566	Identify the renal segments that respond to the following hormones: atrial natriuretic factor, parathyroid hormone, anti diuretic hormone and aldosterone, and the effects of each hormone on the segments.
SOM.MK.1.BPM1.1.CPR.3.HCB.0567	Describe the role of the renin-angiotensin-aldosterone system (RAAS) in Na ⁺ and blood volume homeostasis.
SOM.MK.1.BPM1.1.CPR.3.HCB.0568	Describe the role of the renin-angiotensin-aldosterone system (RAAS) in blood pressure regulation.
SOM.MK.1.BPM1.1.CPR.3.HCB.0569	State the role of antidiuretic hormone (ADH) in the regulation of the function of the collecting duct.



Recommended Reading

Histology – A Text and Atlas

Pawlina 9th Edition

Chapter 20: Urinary Systems

Pages: 768 – 813

[Urinary System | Histology: A Text and Atlas: With Correlated Cell and Molecular Biology, 9e | Premium Basic Sciences | Health Library \(oclc.org\)](#)



Overview

- **Parts of the Urinary System**

- **Nephron**

- **Components:**

- **Proximal Straight Tubule**

- **Loop of Henle**

- **Associated Structures: Vasa Rectae, Renal Interstitium**

- **Distal Straight Tubule**

- **Distal Convoluted Tubule**

- **Associated Structures: Juxtaglomerular Apparatus**

- **Connecting Tubule**

- **Collecting Tubule**

- **Excretory Passages**

- **Minor & Major calyces, Renal Pelvis**

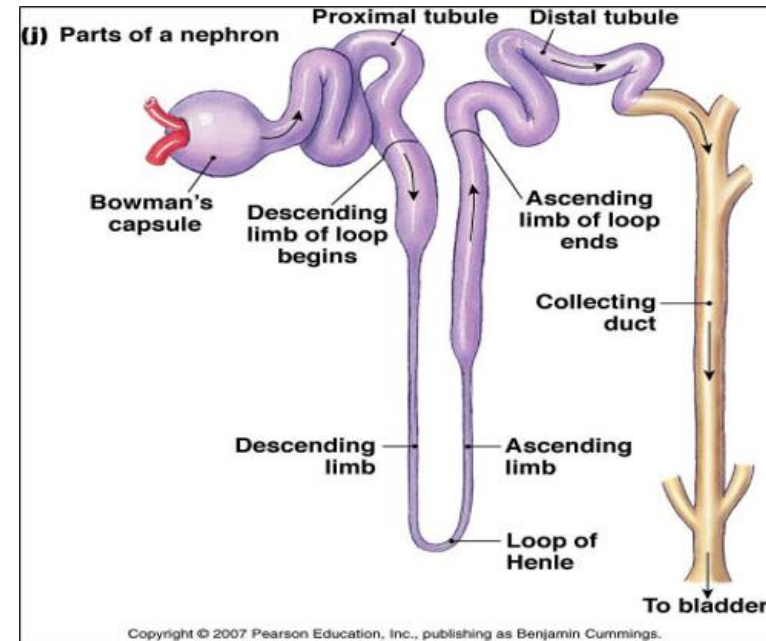
- **Ureters**

- **Urinary Bladder**

- **Urethra**

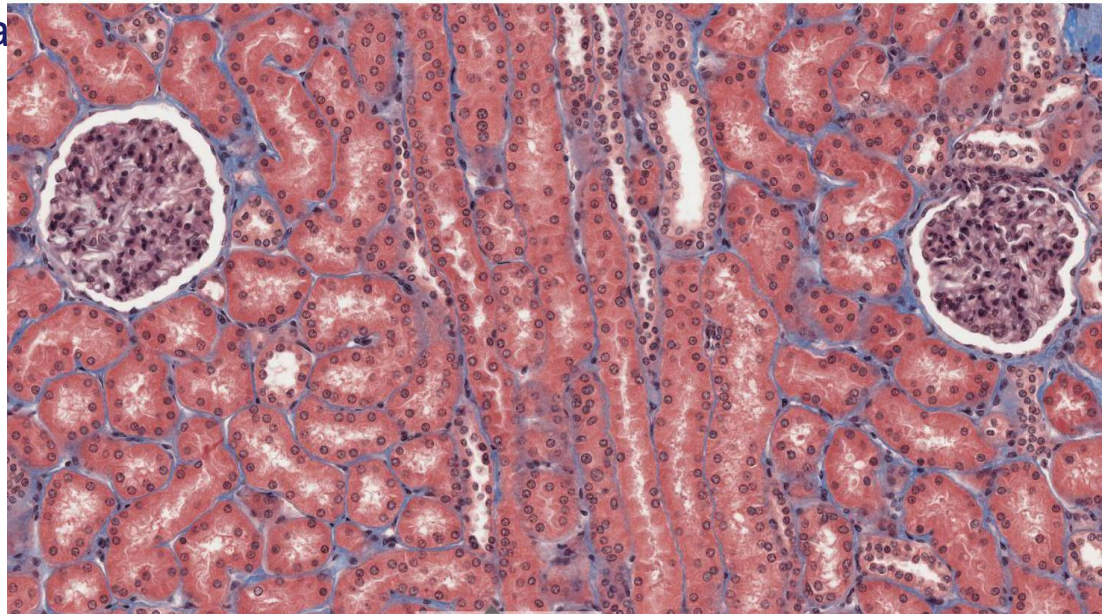
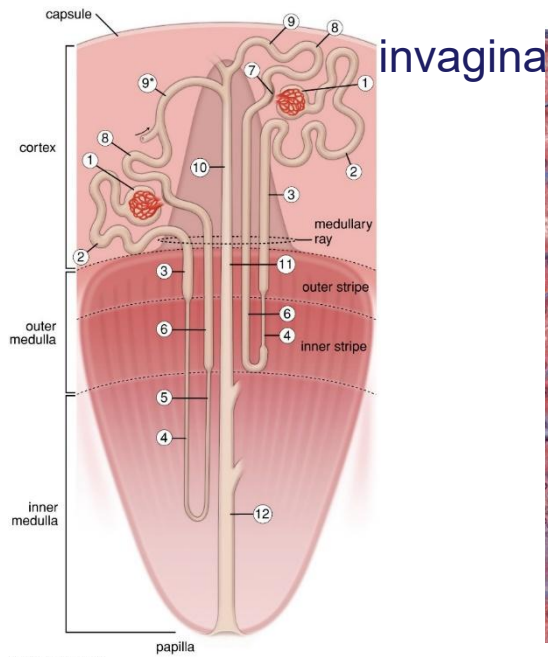
The Renal Nephron

- The functional unit of the kidney.
- **Renal Corpuscle:**
 - **Glomerulus**
 - **Bowman's Capsule**
- **Renal Tubules**
 - **Proximal Convoluted**
 - **Straight/Loop of Henle**
 - **Thick Descending/Proximal Straight.**
 - **Thin descending**
 - **Thin Ascending**
 - **Thick Ascending/Distal Straight**
 - **Distal Convoluted**
- **Uriniferous tubule: Nephrons + connecting tubule.**
Connecting tubule connects the nephron to the collecting duct.



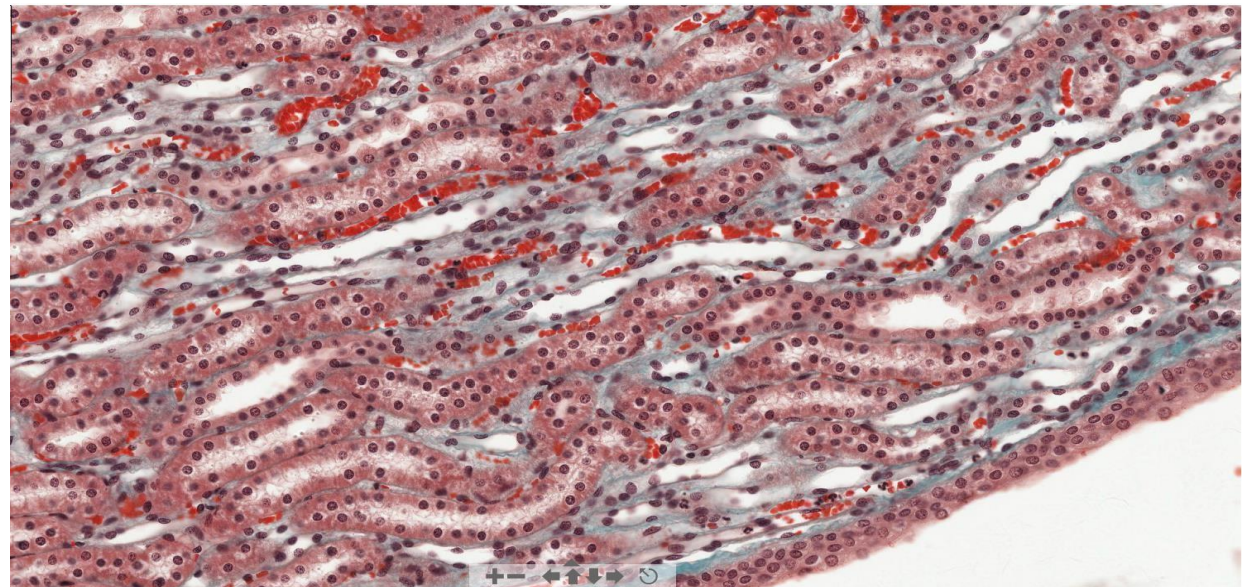
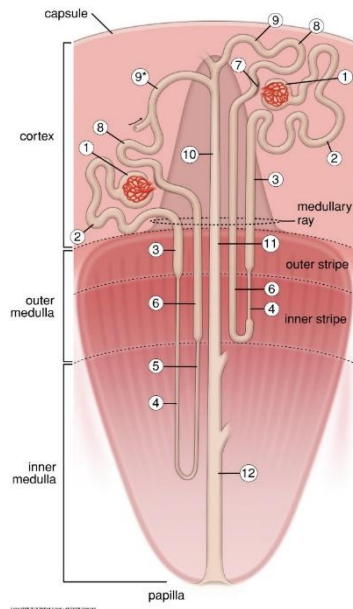
Proximal Straight Tubule

- **AKA Thick descending portion of the loop of Henle.**
- Found in the medullary ray of the cortex and the outer medulla.
- Function :recover any remaining glucose from the proximal convoluted tubule
- Shorter cuboidal cells
 - **Less well-developed brush border.**
 - **Less complex basolateral and lateral interdigitations.**
 - **Smaller mitochondria** that are randomly distributed.



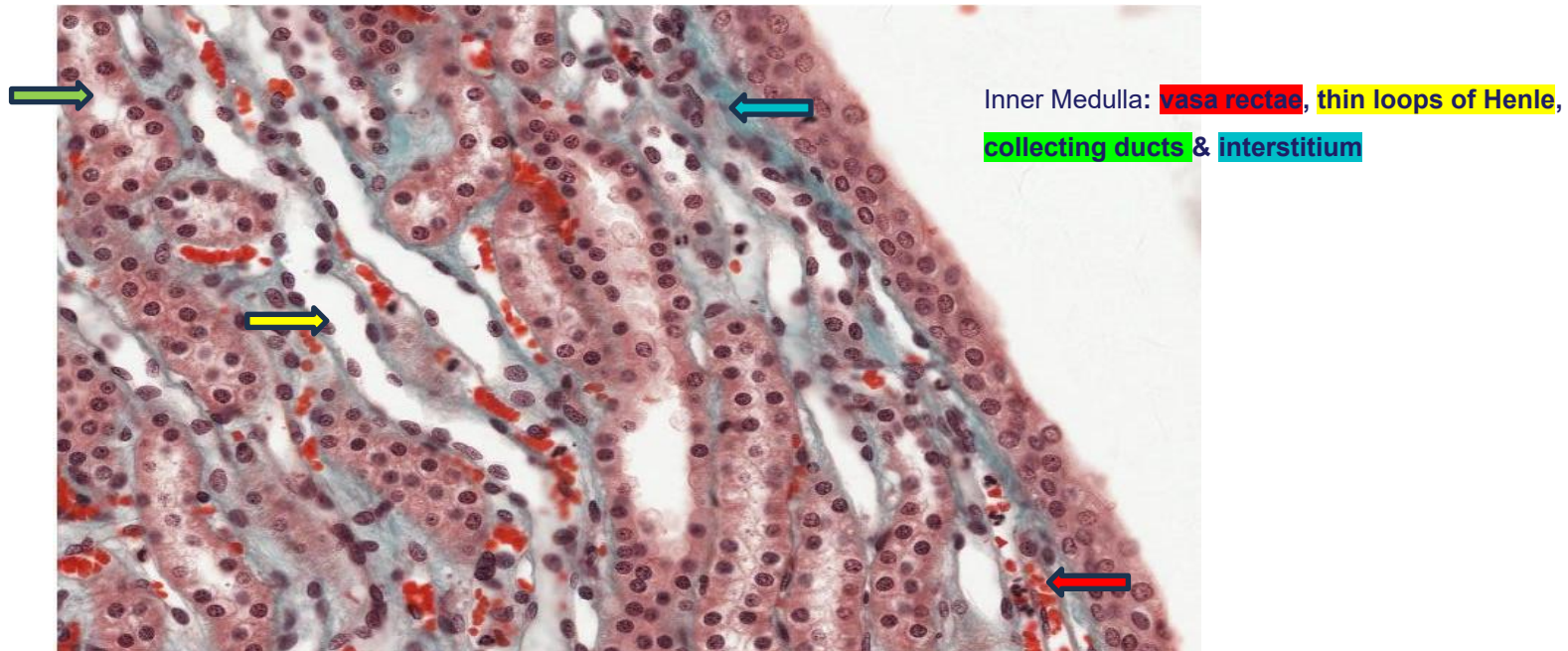
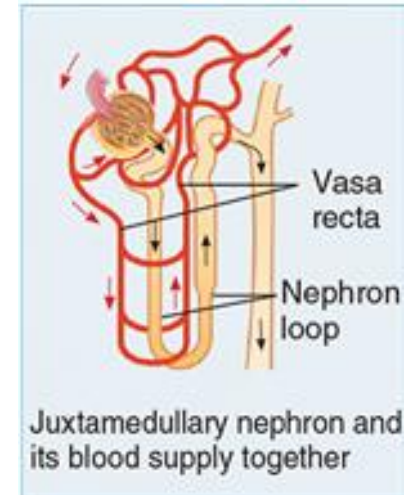
Thin Segment of the Loop of Henle

- Found in both cortical and juxtamedullary nephrons.
- Function: Concentrating tubular fluid via the countercurrent exchange system.
- **Thin descending limb: highly permeable to water** due to presence of aquaporins.
- **Thin ascending limb: Highly permeable to Na^+ and Cl and largely impermeable to water**
- Epithelium: Thin simple squamous in both ascending and descending limbs.
 - May be taller in the cortex VS the inner medulla.



Vasa Rectae

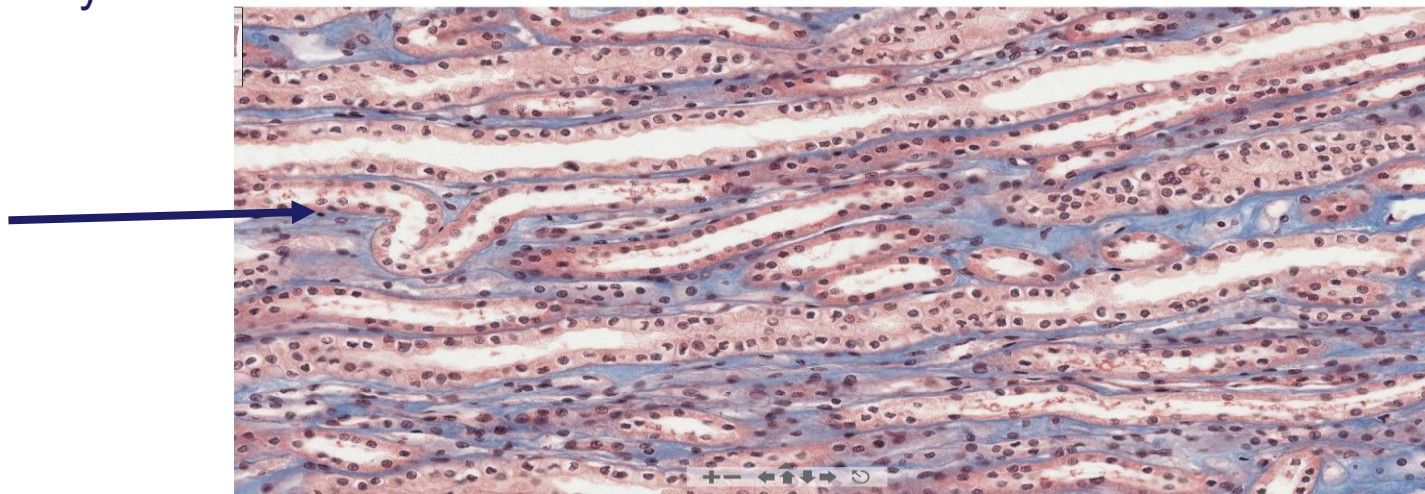
- Arteriole and venule rectae (fenestrated capillaries) forms loops parallel to the loop of Henle and act as **countercurrent exchangers**.
- **Helps maintain osmotic gradient in the medullary interstitium.**



Renal Interstitium

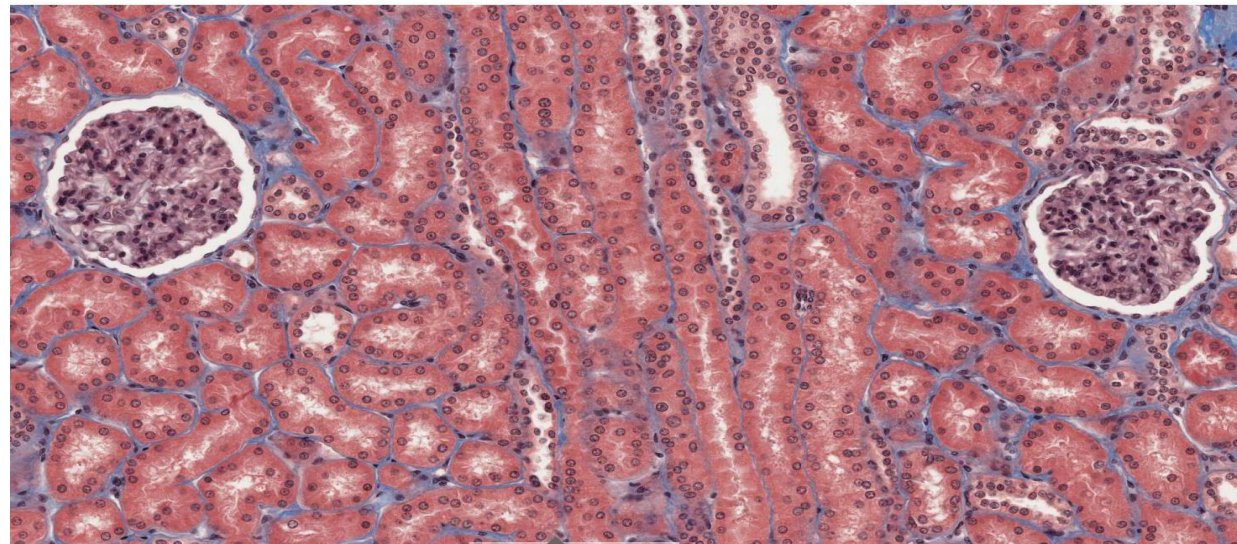
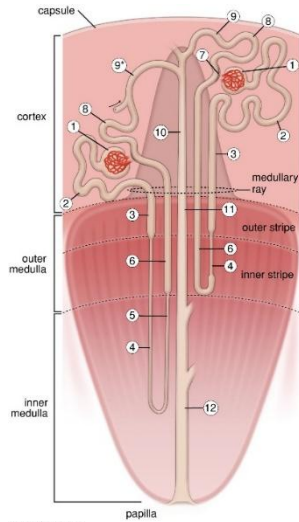
- The connective tissue of the kidney
 - **Surrounds the nephrons and the neuro-vasculature.**
 - **Contains Extracellular matrix (ECM) and cells.**
- **Cortical Cell Types:**
 - Fibroblasts: synthesize and secrete the collagen and glycosaminoglycans of the ECM.
 - Macrophages
- **Medullary Cell Types:**
 - Resemble myofibroblasts

Renal
Interstitium



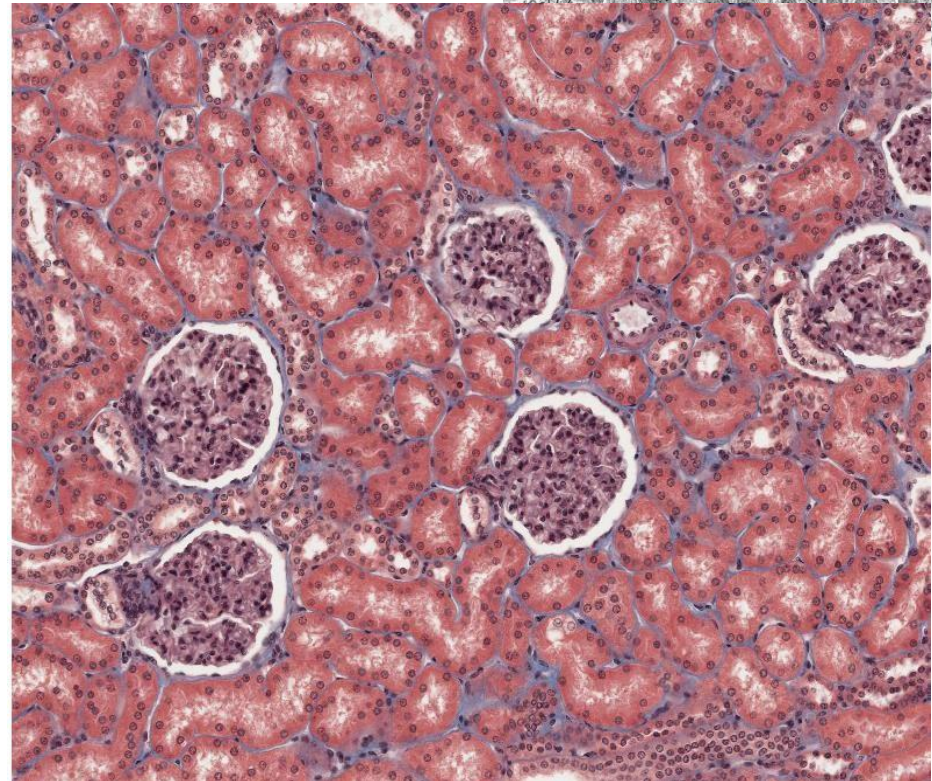
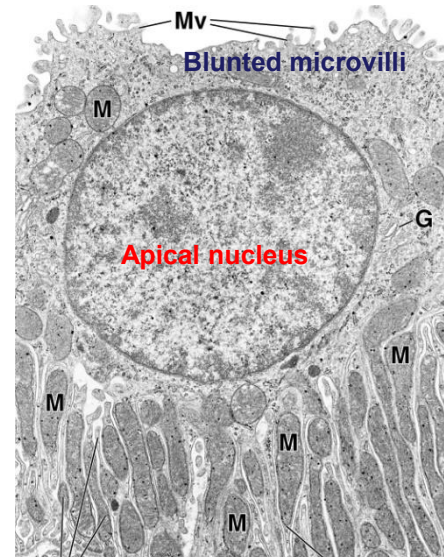
Distal Straight Tubule

- **AKA Thick ascending portion of the loop of Henle.**
- Found in the medullary ray of the cortex and in outer medulla.
- Large cuboidal cells with **blunted brush border**
 - **Nucleus located in the apical portion.**
 - Extensive basolateral plications with numerous mitochondria associated with them.
 - Less mitochondria scattered throughout the cytoplasm.
- **Impermeable to water.**
- Transports ions from the lumen of the tubule to the interstitium via mainly active transport.
 - Reabsorption of other ions: Ca^{2+} and Mg^{2+}



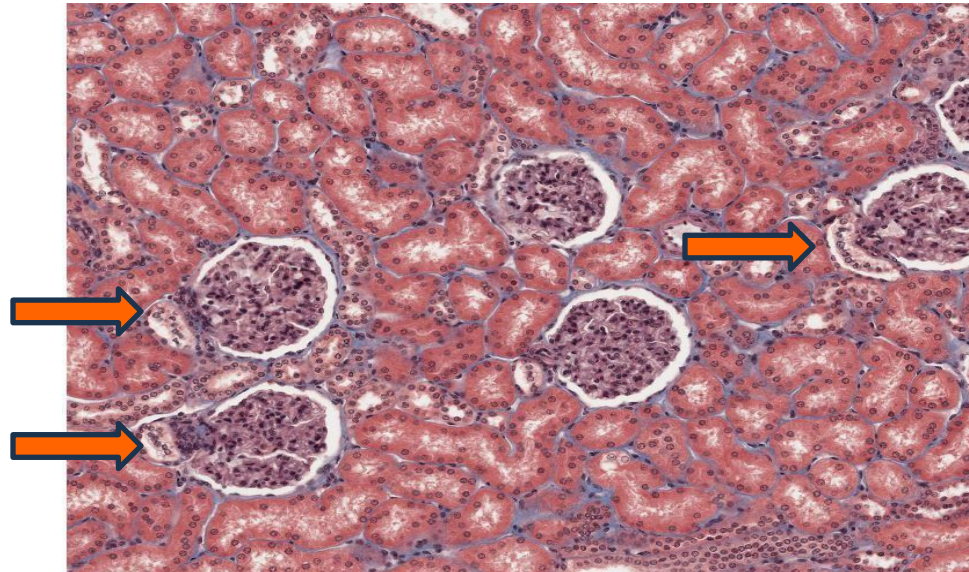
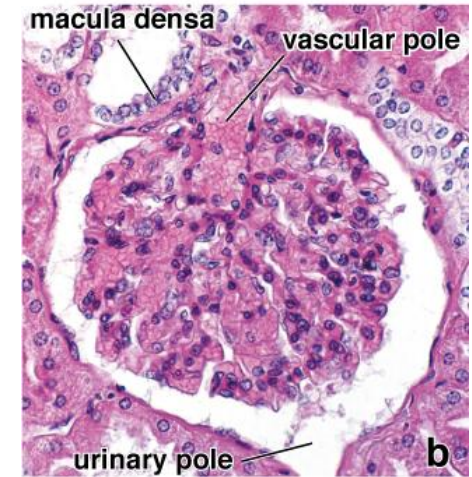
Distal Convoluted Tubule

- Less numerous profile in sections as it 1/3 the length of PCT.
- Simple cuboidal epithelium:
 - Fewer and shorter microvilli – clear lumen
 - Apically displaced nucleus
 - Extensive basal folds and less acidophilic cytoplasm
- Functions:
 - **Relatively impermeable to water.**
 - **Reabsorption of Na^+ and secretion K^+ (highest Na^+/K^+ ATPase activity).**
 - Reabsorption of Bicarbonate and secretion of H^+
 - Secretion of ammonium
 - **Parathyroid regulated Ca^{2+} reabsorption.**
 - Reabsorption of chloride (Cl^-)



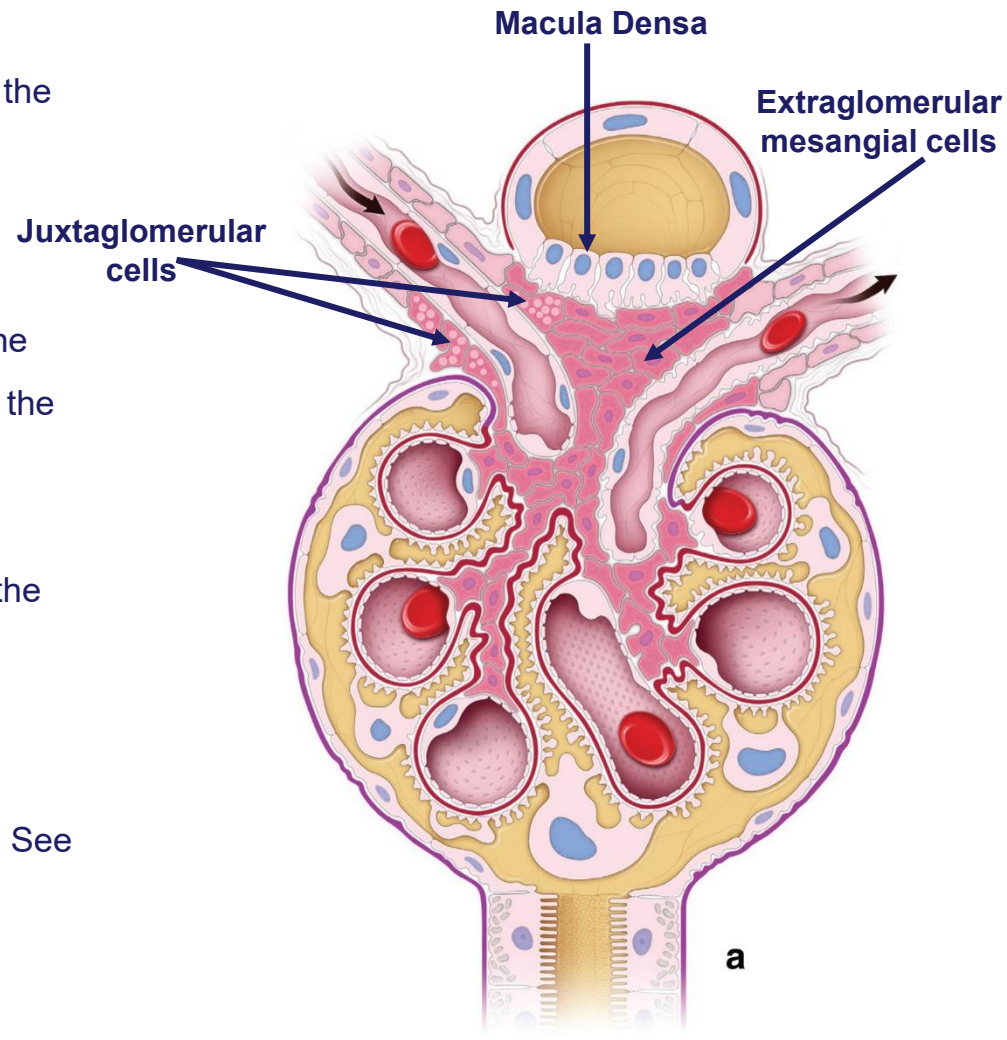
Macula Densa

- **Modified cells** of the distal straight tubule.
 - Narrower and taller cells.
- Located at the **vascular pole**.
- Forms part of the **Juxtaglomerular Apparatus**
 - Regulates blood pressure.
 - Senses changes in Na^+ concentration within the lumen of the distal tubule and sends signals to the juxtaglomerular cells (modified smooth muscle cells located in the afferent arteriole).

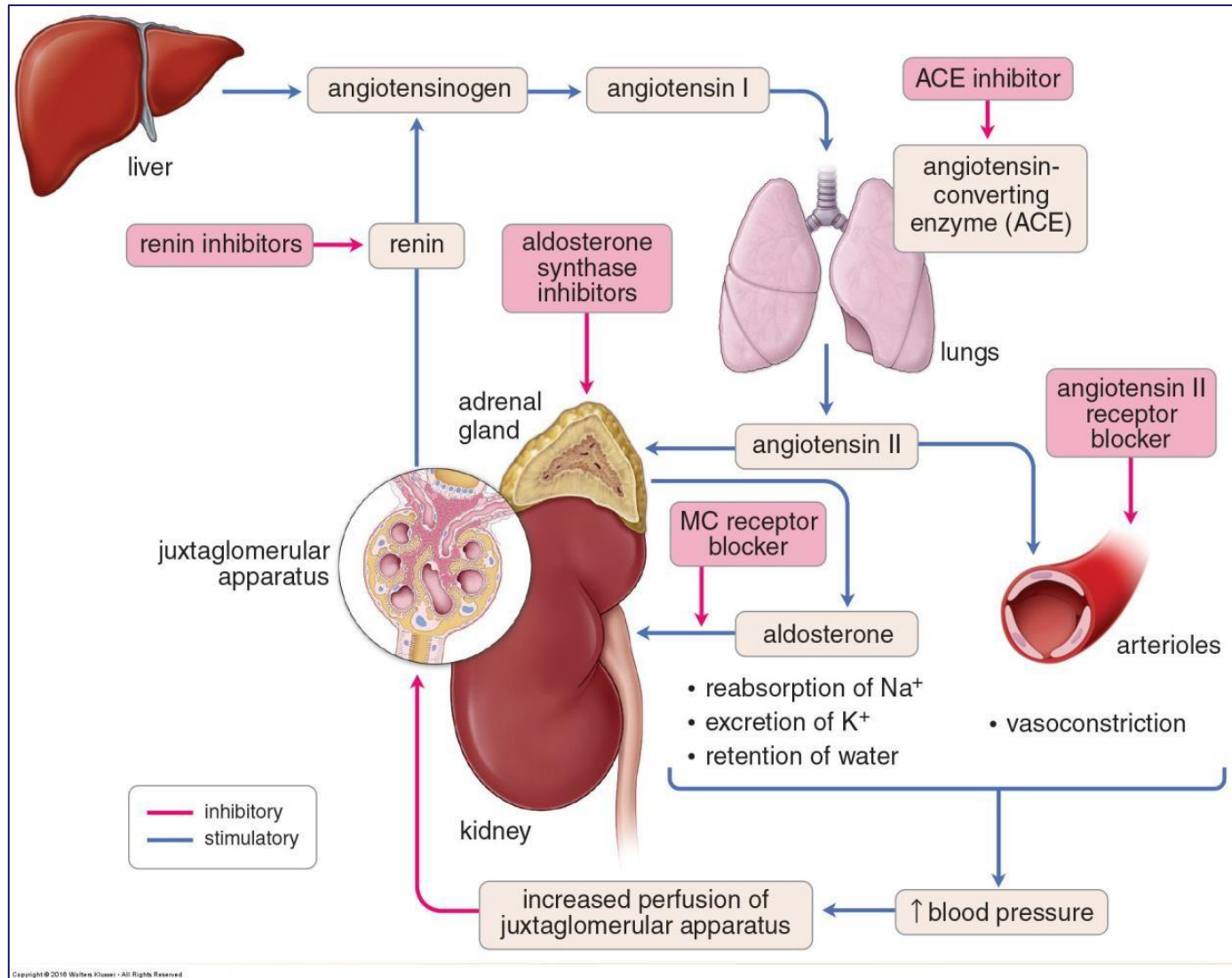


Juxtaglomerular Apparatus

- **Function:**
 - Blood pressure regulation by activating the renin-angiotensin-aldosterone system.
- **Components:**
 1. **Macula Densa (osmoreceptors)**
 - **Monitor Na⁺ concentration** in the tubular fluid and sends signals to the juxtaglomerular cells.
 2. **Juxtaglomerular cells (JG cells)**
 - Modified smooth muscle cells of the afferent arteriole.
 - Senses low blood volume
 - **Produce renin** that cleaves angiotensinogen to angiotensin I. See next page.
 3. **Extraglomerular mesangial cells**
 - Lacis/Polkissen cells

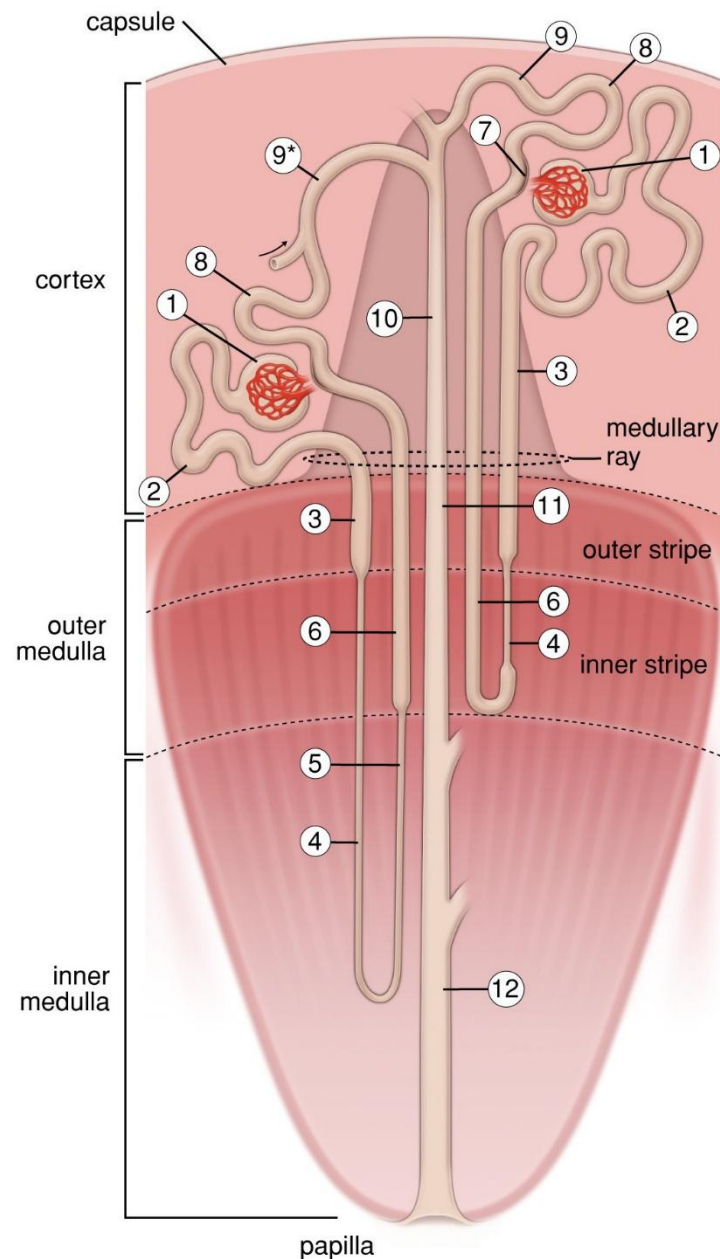


Renin-Angiotensin-Aldosterone System (RAAS)



Connecting Tubule

- **Connects DCT to cortical collecting duct.**
- Cortical nephrons – very short
- Intermediate and juxtamedullary nephrons: longer and arched.
- Gradual transition of epithelium of DCT to collecting duct.
 - Contains cells from DCT and CD (principal cells)
- **Plays important role in secretion of K^+ into lumen – partly regulated by aldosterone.**



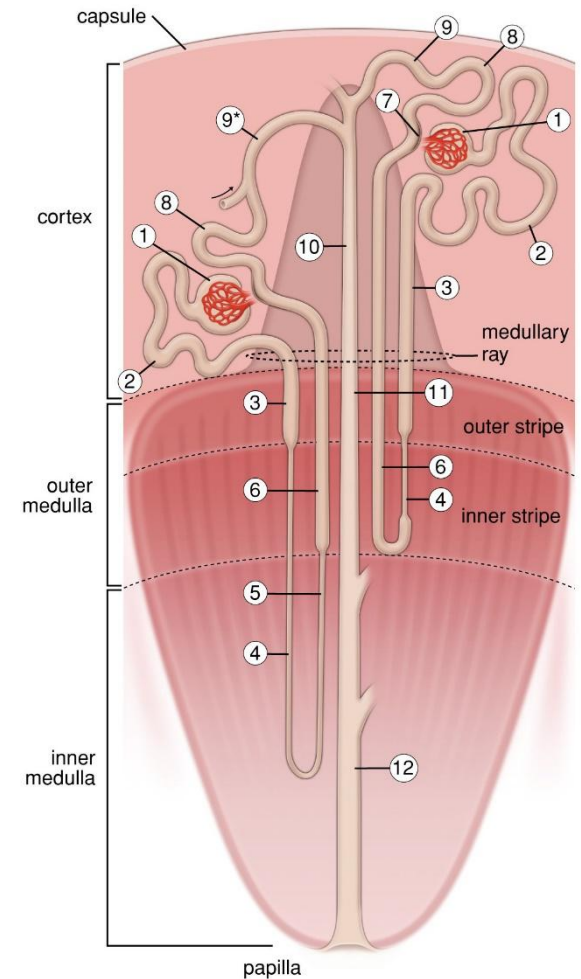
Collecting Ducts

1. Cortical

- Found in the medullary ray of the cortex.
- Connecting tubules lead into cortical collecting duct.
- All nephrons within the renal lobule drain into one cortical collecting duct.
- Lined by simple flattened (squamous-like) to cuboidal epithelium with prominent cell boundaries.

2. Medullary

- The diameter of this duct progressively increases as it enters the medulla.
- Lined by simple cuboidal with a transition to columnar cells.
- Several medullary collecting ducts merge to form papillary ducts (ducts of Bellini).



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Collecting Ducts

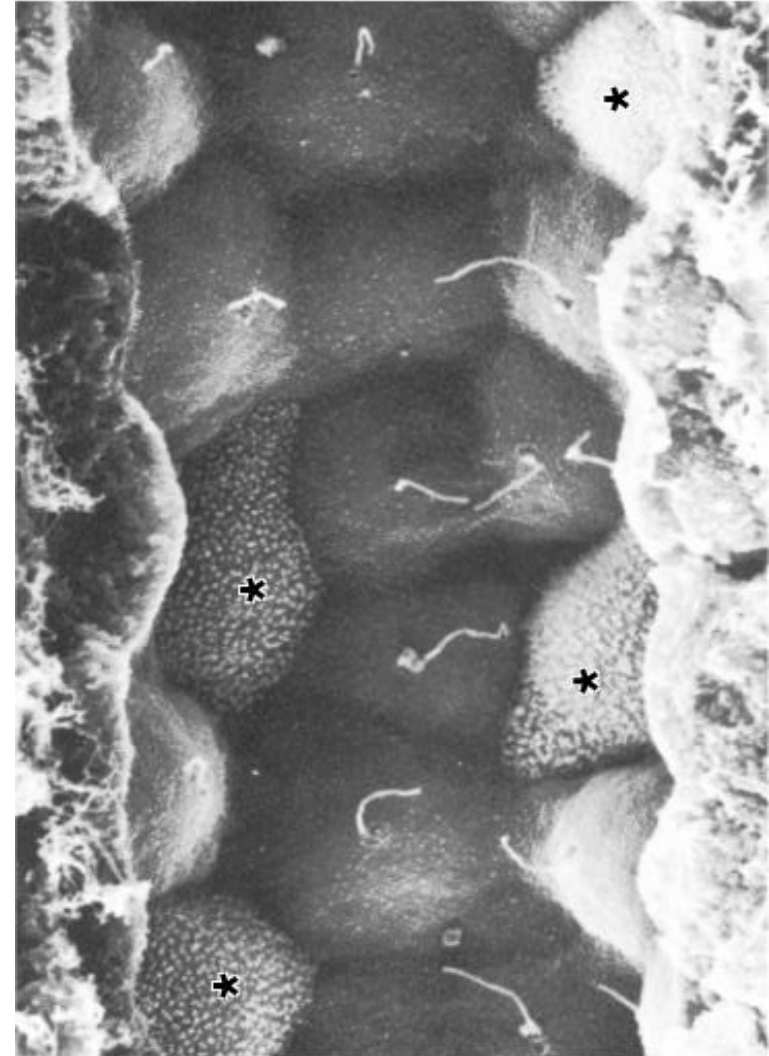
- There are two cell types presents:

1. Light/Principal Cells:

- **Most abundant: pale-staining**
- True basal infoldings
- Single primary cilium: short scattered microvilli
- **Function: water reabsorption.**
 - Abundant ADH regulated aquaporins (AQP-2) channels.
 - Have many receptors for aldosterone.

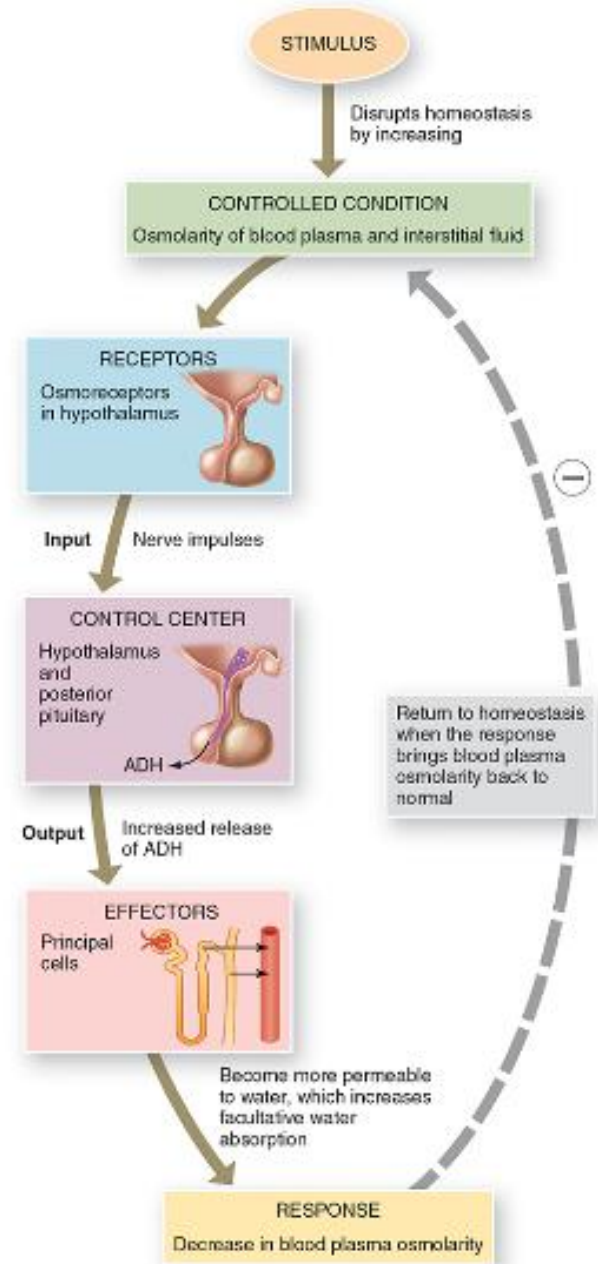
2. Dark/Intercalated cells

- **Fewer**
- Not present in the inner medulla
- Microvilli and micro plicae at apical cytoplasm
- **Function: acid-base homeostasis**
 - Secretion of H^+ (alpha intercalated) or bicarbonate (beta intercalated).

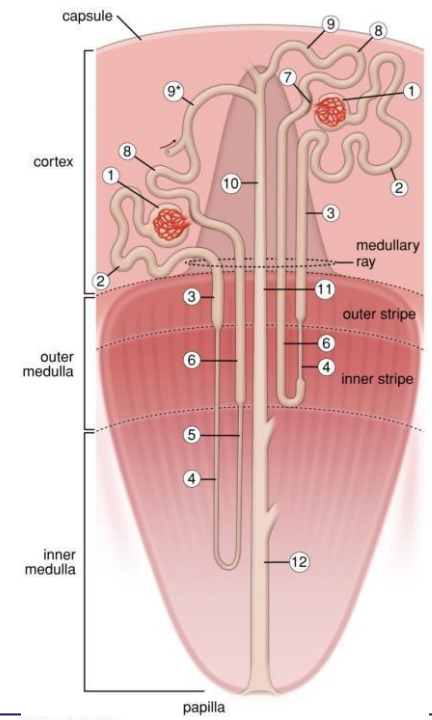
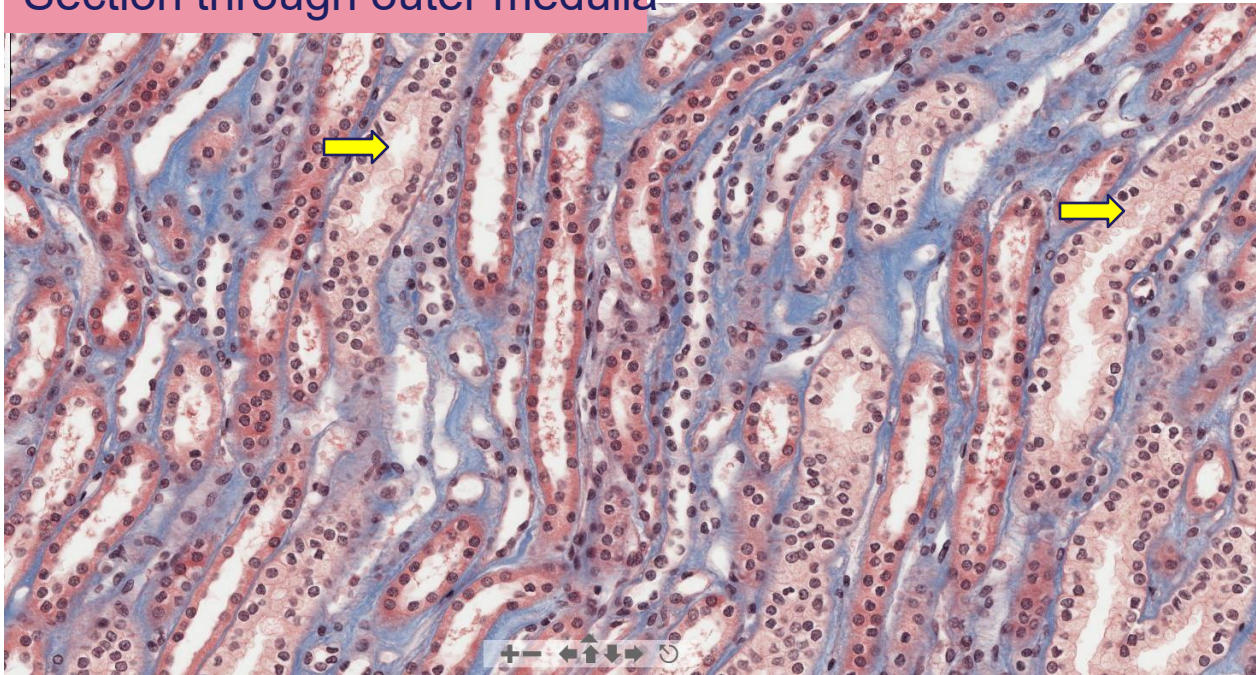


Antidiuretic Hormone

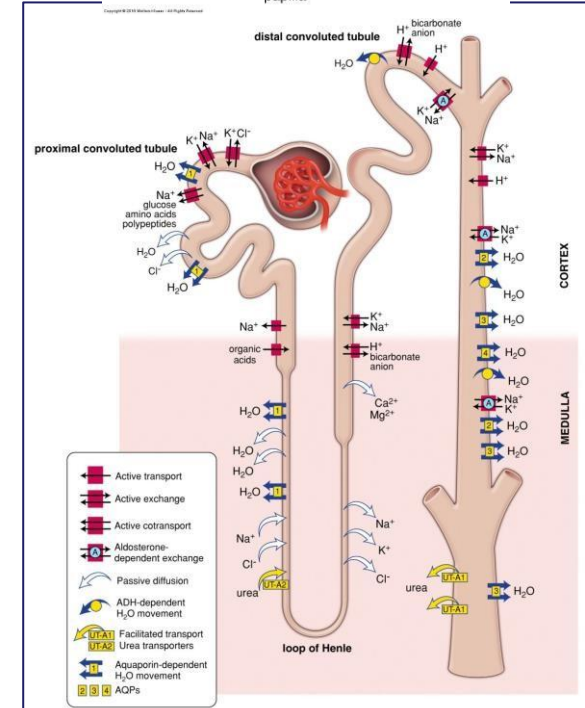
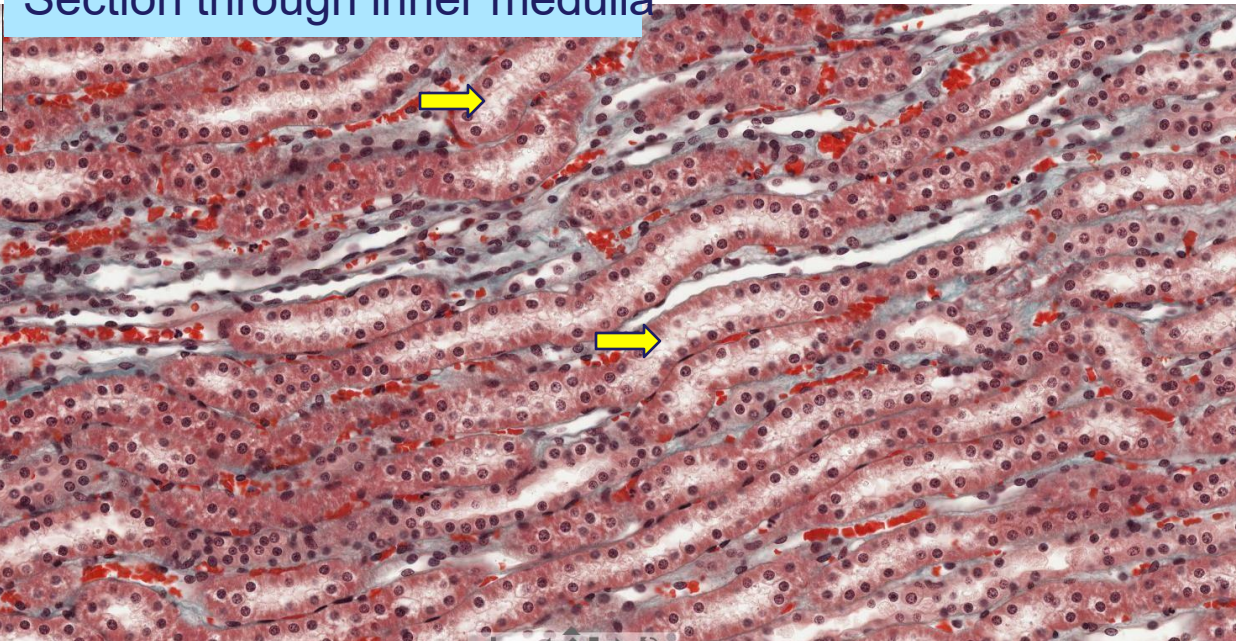
- Produced by the hypothalamus and released from the pituitary gland posterior lobe.
- Stimulated by an increase in plasma osmolality or decrease in blood volume.**
- It increases the permeability of the collecting ducts to water by:**
 - Synthesizing and translocating more AQP-2 channels to the apical cell membrane.
- Causing the increased reabsorption of water, causing the production of more concentrated urine.



Section through outer medulla

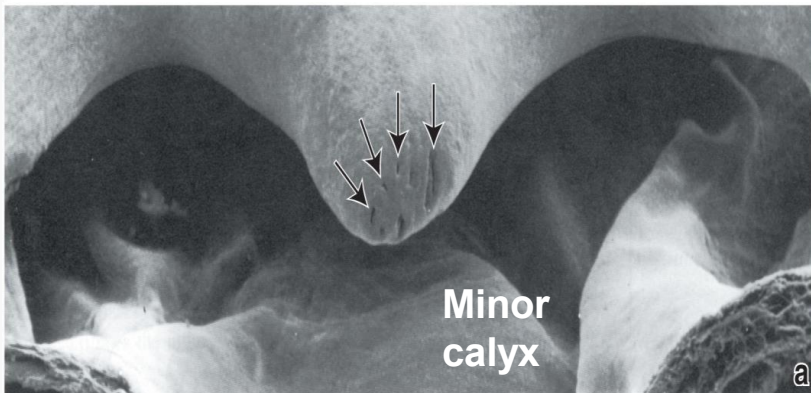
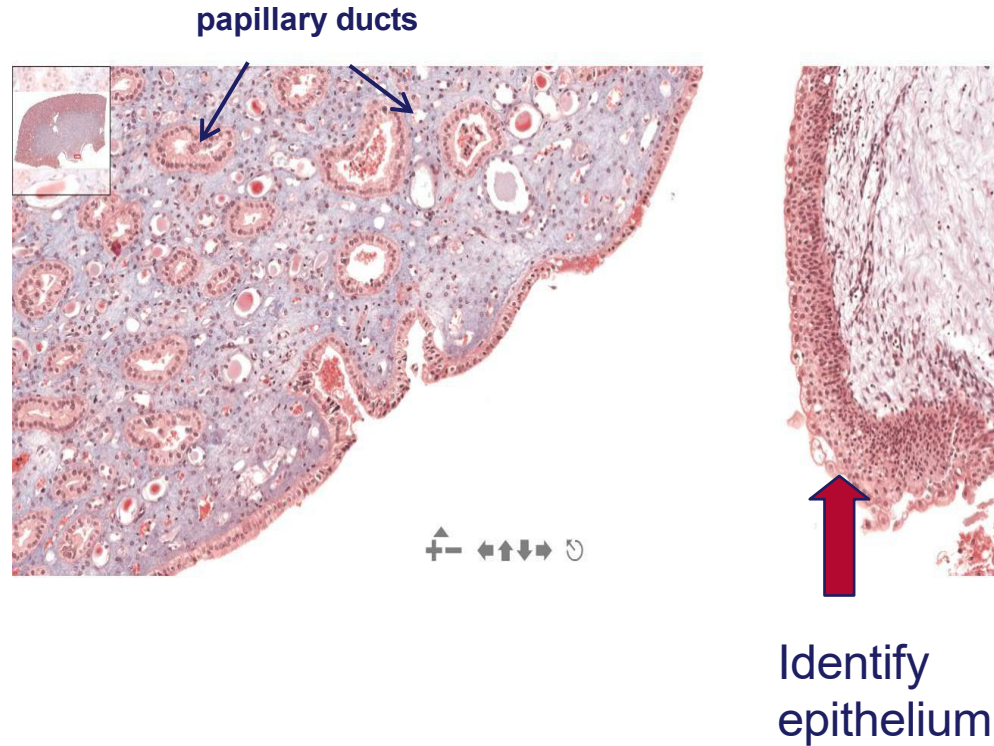


Section through inner medulla



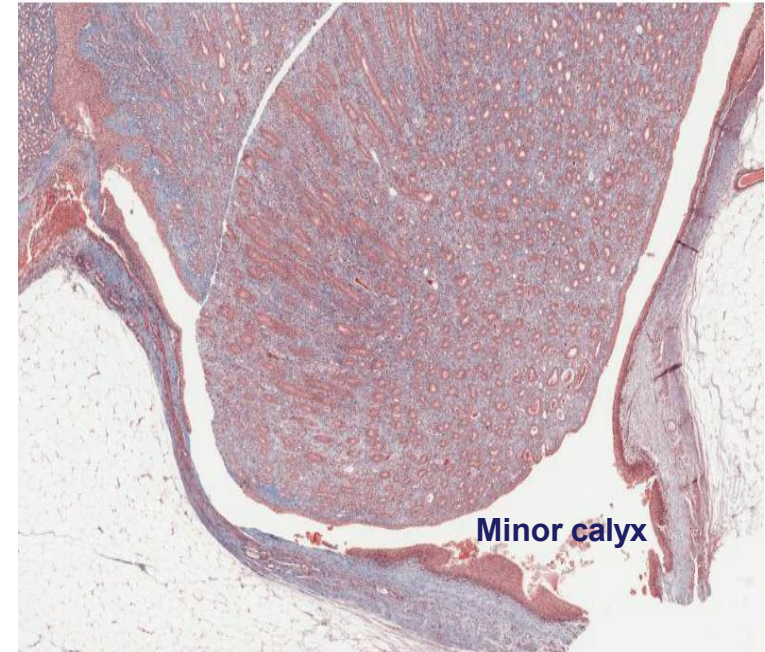
Renal Papilla

- The ducts of Bellini (AKA papillary ducts) empty via the area cribrosa at the apex of a **renal pyramid** which is called the **renal papilla**.
- Also marks the site of the beginning of transitional epithelium.



Excretory Passages

- Functions:
 - Transit and storage of urine.
- Include:
 - **Minor Calyx** (8-18):
 - Cup-shaped and surrounds a single renal papilla.
 - **Major Calyx** (2-3):
 - Each receives urine from 3-6 minor calyx.
 - **Renal Pelvis:**
 - Expanded upper end of the ureters that lies within the renal sinus.
 - **Ureter**
 - **Urinary bladder**
 - **Urethra**



Structure/features:

- **Mucosa**
 - **Urothelium** AKA Transitional epithelium
 - Lamina propria.
- **Muscularis**
- **Adventitia/serosa**

Urothelium

3 layers of cells.

1. Superficial:

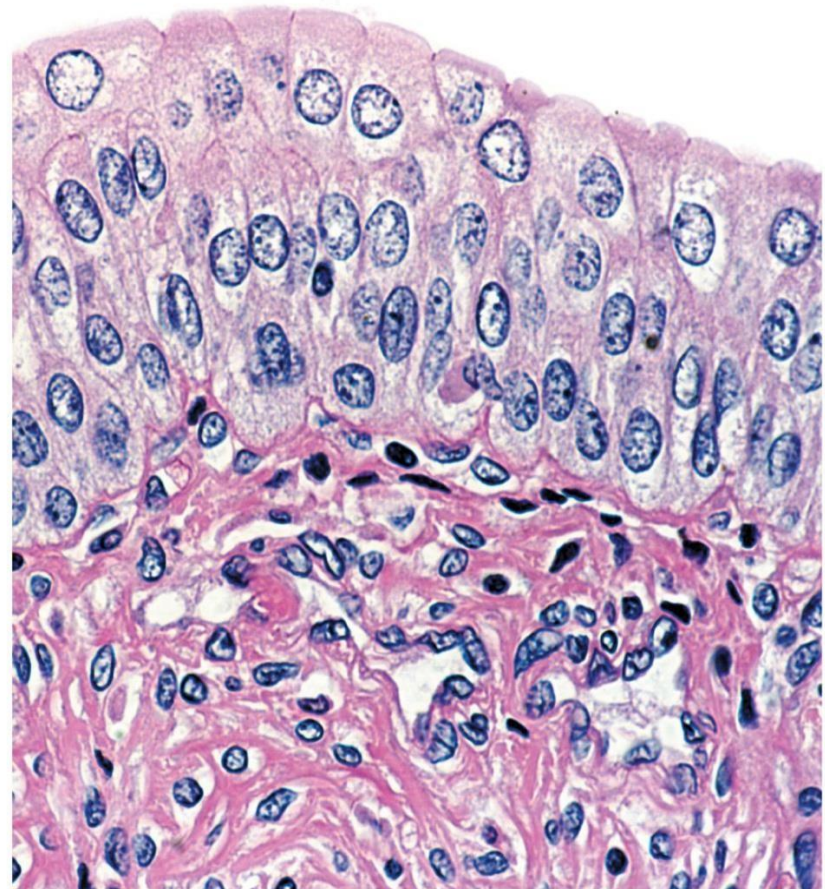
- Single layer of dome-shaped (umbrella) cells.
- The shape of the cell depends on the filling state.
 - Abundant tight junctions
 - Impermeable to urine.

2. Intermediate:

- Pear shaped cells: abundant desmosomes
- Allows for epithelium to transition.

3. Basal Layer:

- Stem cells

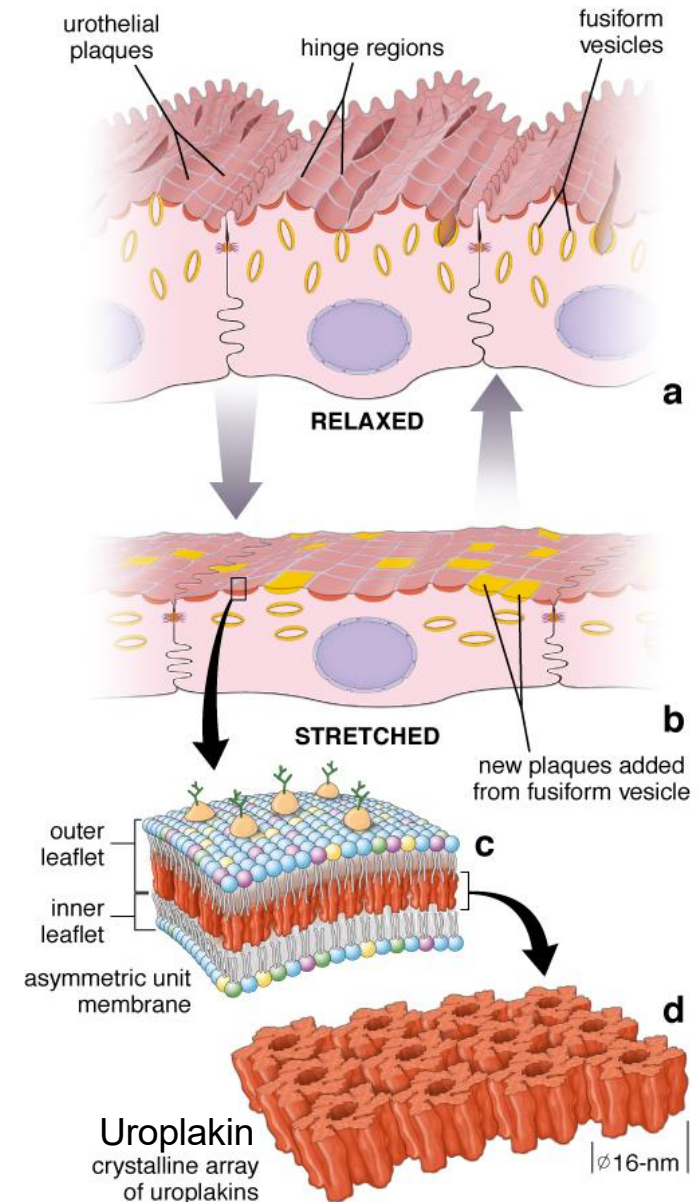
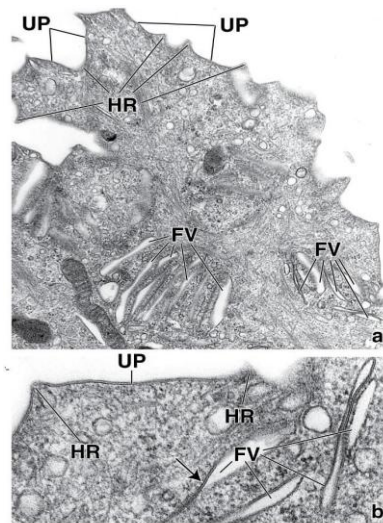


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Superficial Cells – Special Modifications

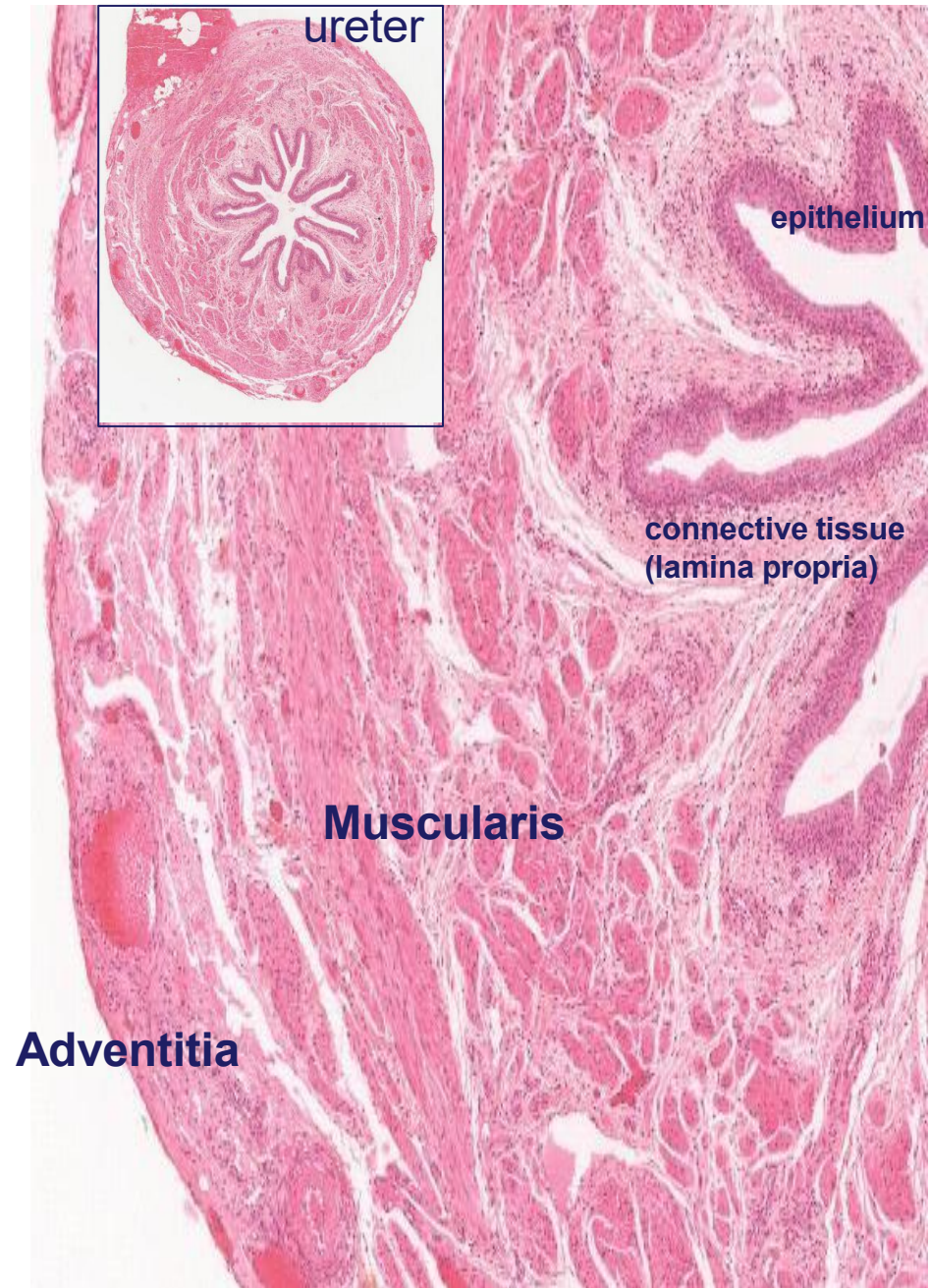
- **Urothelial plaques:** Covers the apical plasma membrane
 - Contains crystalline transmembrane proteins called uroplakin which makes the plaque impermeable to water, urea, protons.
- **Hinge Regions:** Narrow area that separates the plaques.
 - Contains all other non-plaque proteins such as receptors and channels.
- **Fusiform Vesicles:** flattened elliptical vesicles
 - In an empty bladder they are vertically oriented.
 - In a full bladder most vesicles fuse at the hinge region and some become arrange parallel to the apical membrane.

UP- urothelial plaque
FV- fusiform vesicles
HR- hinge regions



Ureter

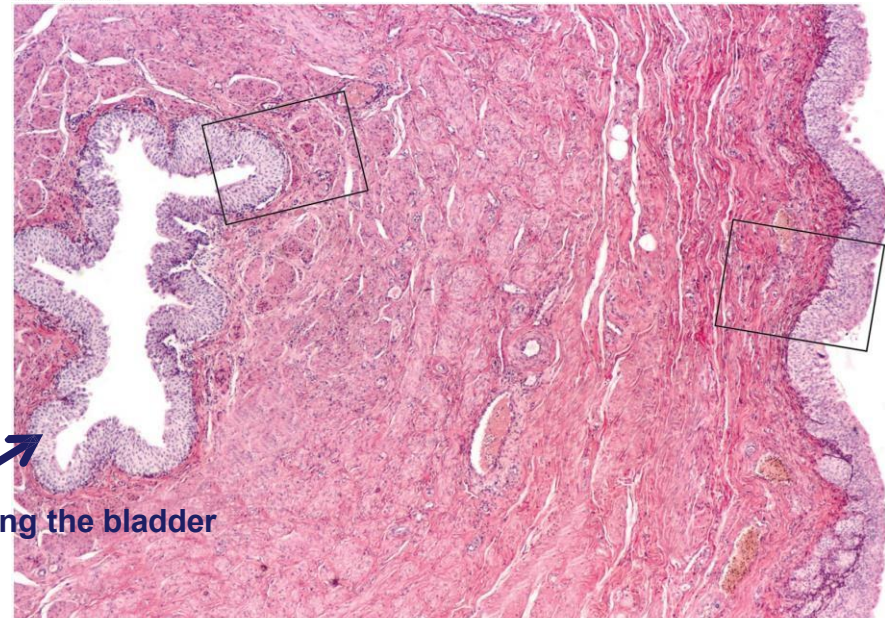
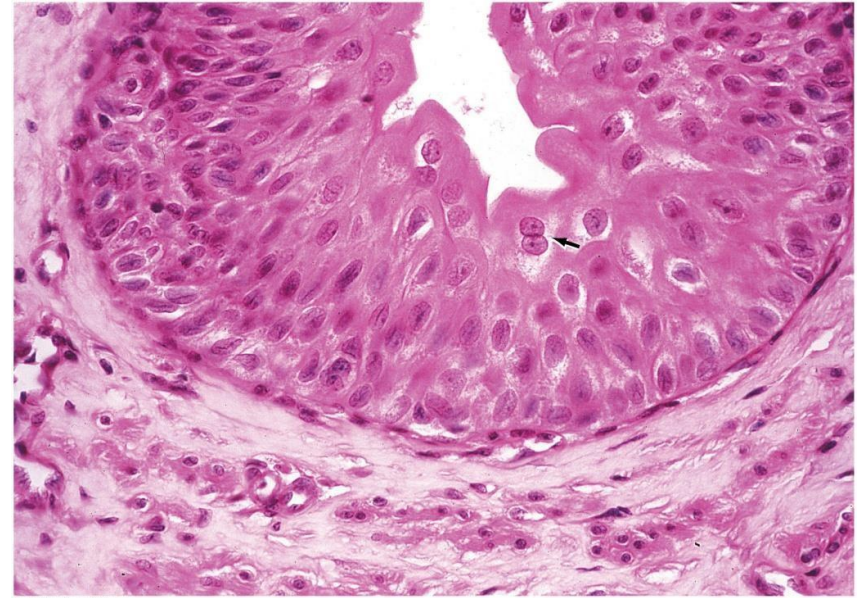
- The conduit between the renal pelvis and the urinary bladder.
- **Mucosa:**
 - Urothelium + Lamina Propria
- **Muscularis:**
 - 3 layers of smooth muscle
 - Inner Longitudinal
 - Middle Circular
 - Outer Longitudinal (only present in the distal 1/3)
- **Adventitia:**
 - adipose tissue, vessels and nerves.



Urinary Bladder

General organization:

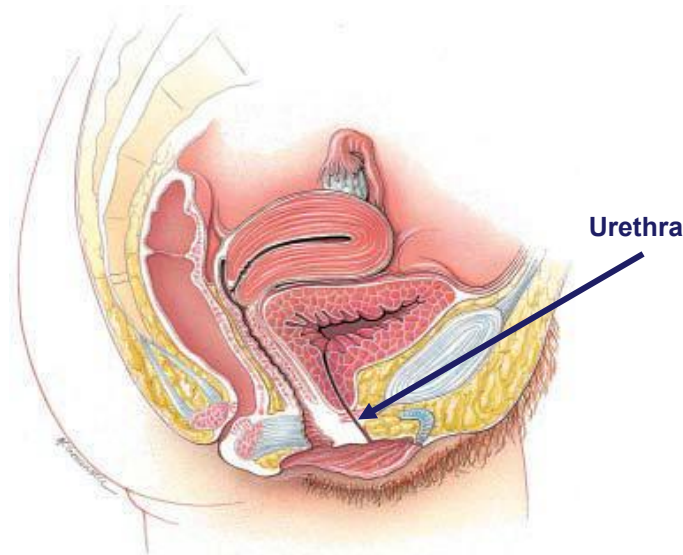
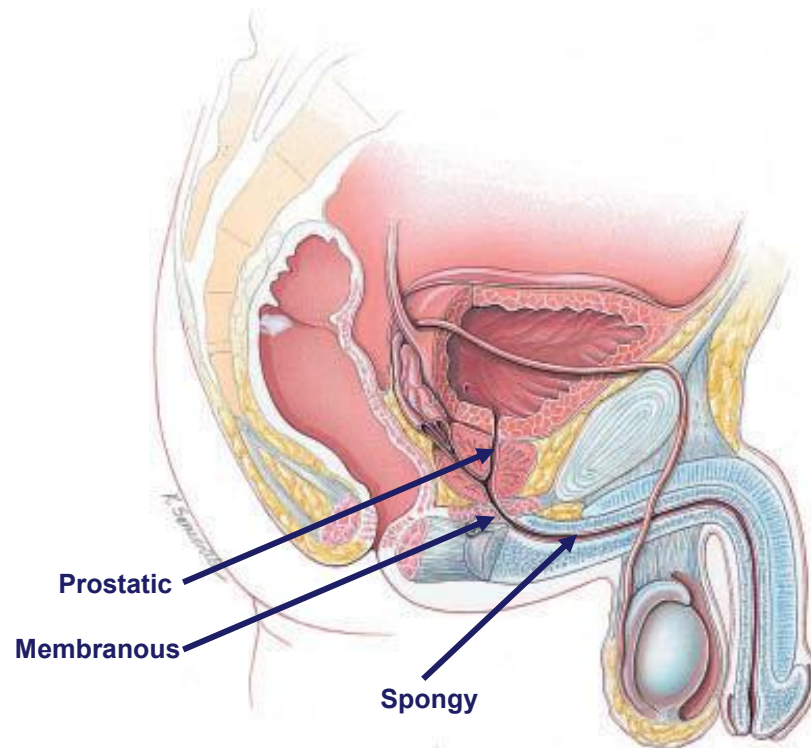
- **Mucosa**
 - Urothelium + fibroelastic CT
- **Muscularis (Detrusor muscle)**
 - Thick layers of smooth muscle arranged in **an inner longitudinal, middle circular and outer longitudinal layer (oblique orientation of muscles)**.
 - Forms the involuntary internal urethral sphincter around the opening of the urethra.
- **Adventitia / serosa**



Ureter entering the bladder

Urethra

- Conveys urine from bladder to exterior through the external urethral orifice.
- **Male Urethra:**
 - **Prostatic:** transitional epithelium
 - **Membranous:** transitional epithelium
 - **Spongy:** lined by pseudostratified columnar epithelium except at the distal end where it is lined by stratified squamous.
- **Female Urethra:**
 - Short: high risk of urinary infections
 - **Initial portion:** transitional epithelium
 - **Terminal portion:** stratified squamous epithelium.





Overview

- **Parts of the Urinary System**

- **Nephron**

- **Components:**

- **Proximal Straight Tubule**

- **Loop of Henle**

- **Associated Structures: Vasa Rectae, Renal Interstitium**

- **Distal Straight Tubule**

- **Distal Convoluted Tubule**

- **Associated Structures: Juxtaglomerular Apparatus**

- **Connecting Tubule**

- **Collecting Tubule**

- **Excretory Passages**

- **Minor& Major calyces, Renal Pelvis**

- **Ureters**

- **Urinary Bladder**

- **Urethra**